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RESOURCE DRIVEN URBAN METABOLISM

How can metabolic scaling affect urban design?

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Abstract. City scale projects are complex multivariable problems and have previously been addressed using a variety of organisational principles, whether it be the infrastructural grid, used by Ludwig Hilbersheimer in his project "Hochhausstadt" or the spinal organisation of the Tokyo Masterplan done by Kenzo Tange and the Metabolist Movement. This project strays from traditional methods of structuring a city and investigates a novel self-engineered anticipatory model, which focuses on the use of generative and genetic algorithms to develop a new associative system to develop coastal cities in arid climates. The system functions as a negative feedback loop, analysing existing conditions, and by a series of mathematical functions, projecting the new growth patterns for major components of a city, such as building envelopes, road networks, canal networks and public space distribution as a result of the cities internally generated resources.

Keywords. Computational design; generative & evolutionary design; tooling; city modelling; urban metabolism.

1. Introduction

An IPCC study indicates that desert climates are expanding (IPCC, 2007), which points to current problems of living in desert climates as being a pressing matter. This project tries to mitigate some of these immediate problems by proposing a new development model for Qatar, which has its high CO^2emissions (The National, 2011), high population growth of 12.6% per year (WolframAlpha, 2011) and very little to no

access to freshwater. There are no rivers or lakes, and the country has an annual rainfall below 75mm per year (Gaisma, 2012). The rainfall, and the small amount of surface water this creates, is almost the only access to freshwater for the population of Qatar, which has made them heavily dependent on the energy intense process of desalination since the 1950's for their freshwater needs (Al Mohannadi, 2010).

2. Precedents

Developing new design strategies for urban developments is not a new idea, and there are two specific projects that are beneficial to analyse in relation to their drivers for urban growth in order to describe how the methods utilized in this research project separates itself from its precedents.

In 1924 Ludwig Hilbersheimer proposed a theoretical project named "Hochhausstadt" (– "Skyscraper city") where the city is expressed as an infrastructural grid fitted with architectural program (Aureli, 2011, pp. 13-16). Having the grid as the underlying structure for the city makes the process of urbanization a predictable task, and the outcome very homogeneous. The project is seen as a first attempt to enable people to model and comprehend the complex relations, of an infinite city without the use of heavy computing, because of the modular logic that has been developed for the design proposal.

In 1960 Kenzo Tange proposed his theoretical plan for Tokyo, a plan where the city is organized linearly around a central axis (a road) from which secondary roads branch off (Lin, 2007). Even though the organization of Tange's plan differs from Hilbersheimer, the generated morphology is equally as predictable, quoting Peter Smithson:

> "Having a central axis of transport, from which all network originates, will require all vehicles to cross the central axis at some point on its way from point A to B, even though they have no business there." (Lin, 2007)

This reasoning suggests that people living in this city, would be more likely to live along the central axis in order to minimize their driving distance, and thereby create a long thin city.

The above examples indicate, that using traditional means of urban growth, such as infrastructure will create predictable urban morphologies. The intention of this project is to create a novel growth model, by using a resource-based driver for the generation of a city, where you cannot prescribe the resulting morphology.

3. Method

This section will first describe how metabolic scaling was used to dimension building envelopes and different networks of the city, then how genetic algorithms

were used to optimize the branching strategy for a canal network (elaborated in section 4.2).

3.1. METABOLIC SCALING

The city is an incredibly complex system made up of many different variables, both visible and invisible. When looking at cities as single living entities, they seem to posses a metabolism like the one inherent in all living systems.

> Biological metabolism functions through surfaces and branching networks, and there has been a century or more of research into the mathematical parameters common to all living forms. In cities too, the operation of metabolism occurs through urban surfaces and urban networks, and is a remarkable similarity between the mathematics of biological metabolism and urban metabolism. (Weinstock, 2011)

Bettencourt et al. describes, in the paper "Growth, Innovation, Scaling and the pace of life in cities" (Bettencourt et al., 2007), how urban- and metabolic systems have similarities, through a series of mathematical functions. However it is summarized that one power law function can determine the makeup and subsequent scaling of any network or mass sensitive system:

$$Y = Y_0(M^b) \tag{1}$$

Where Y is some observable characteristic, Y_0 is a constant related to that characteristic. M is the mass of the object in question, and b a multiple of ¼ (Brown and West, 2000). This formula was used to determine the footprint (Y) and height (M) for the building envelopes within the city proposals. Y_0, a constant relative to the footprint, was determined by using a "parent" building; where the height and footprint was known, so that it was possible to isolate Y_0 for further use. The parent building was used to test the theoretical concept, saying that a mouse mass is metabolically related that of an elephant, could also be applied to architecture.

Another formula was used to dimension other components of the city, which is presented in the paper: "*The fractal Lung: Universal and species related scaling patterns.*" The formula is as follows:

$$S_z = S_0(\alpha^z) \tag{2}$$

The formula doesn't take mass into consideration, but the depth of a given network. Using a tree as a model: S_z is the size of a branch at the "z" depth level (length or width). S_0 is the size of the trunk, and α^z is a constant where $\alpha < 1$. In

the case of most natural systems where α is about .75, interestingly complies with the rules for mass based scaling (Nelson et al., 1989). The use of this formula was very wide spread throughout the project, but was mainly used to define length and width of the canals and roads in the city. How this worked will be further elaborated in section 3.2, where the generative strategy for the canal network will be explained.

3.2. GENERATIVE AND GENETIC DESIGN

The main component of the project is a generative algorithm, meaning it is generating networks and entities over time (i.e. growth over time). Within the algorithm there are different sub systems that have individual rates of expansion, which will be controlled by an embedded control system (will be further elaborated in section 4.1) that will adjust the different growth rates to meet different criteria, for instance there will never be more people living in the city than it can sustain. Two of the sub systems have been genetically optimized, using a self-made genetic algorithm (Flake, 1998, pp. 339–360). The algorithm searches through a given morphospace (Goodwin, 1994, pp. 112–114) and evaluates the different outcomes based on a fitness formula.

To give an example, the city will have a canal network that branches out through the city (elaborated in section 4.1). The algorithm was developed based on the principles for L-systems and fractals (Flake, 1998, pp. 77–92).

The canal network has a few different parameters that can vary; the branching style (STL = short to long, or LTS = long to short), branching angle and meandering angle.

The branching style (STL or LTS) refers to the length of the different segments, which is defined by the metabolic scaling formula (2). Meaning if the first segment of canal (the trunk) is long then the following segments will get shorter (LTS), and vice versa if you start with a short(trunk) segment (STL).

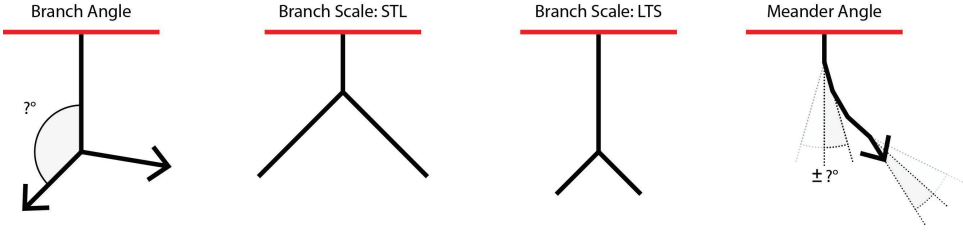


Figure 1. The different branching parameters.

The algorithm was designed to search for a optimal branching strategy for the canal network, the goal (fitness) for the algorithm was to create the network that would have the lowest value according to the fitness formula(3):

$$\text{Fitness Value} = \text{Length of network} / \text{total covered area} \quad (3)$$

It creates an initial population by stochastically arranging values from the parameters within the morphospace. More generations were created enabling or combining multiple parameters; this was to see the effect of varying different parameters. These initial populations are characterised as being the "gene pool", consisting of different parameter sets, i.e., morphospaces. This gene pool is then utilized for further genetic testing by cross breeding different individuals to see what would make viable solutions according to the fitness formula (3). Surprisingly the results from one of the initial populations where the effects of meandering were tested, not where cross breading was introduced, proved to be the fitter strategy. Bellow is an image of the fittest strategy and its associated values.

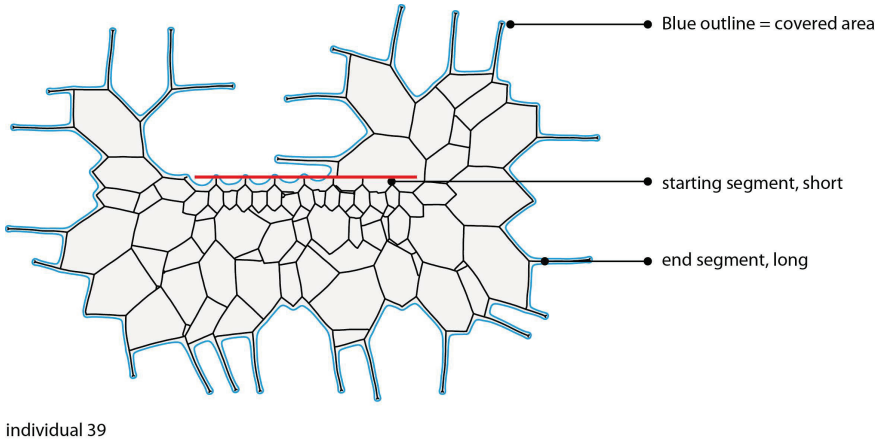


Figure 2. The fittest branching strategy and explanations of how data was derived.

Table 1. Data set for fittest branching strategy.

	Branching strategy	**Network length**	**Branching angle**	**Covered area**	**Meandering angle**
Figure 2	STL	115484,340	45/45	28257424,155	2

4. The Project

This section of the paper will describe some of the components that constitute the possible outcomes of the project, and how the city was enabled to grow as a result of a constant feedback loop between its internally generated resources and its current population. A city requires numerous different resources to sustain it self, however in order to test our proposed methodology, it was decided to focus on water; mitigating the issues of access and acquisition.

4.1. THE WATER STRATEGY

There will be a salt-water canal system dispersed throughout the city. The canals will be fitted with a special railing system that will function as a water productive surface, by being coated with a hydrophilic/hydrophobic polymer that was developed by a group of biomimicry researchers at the University of Sydney. The polymer will enable the system to withdraw water from the relative humidity in the air (Thickett et al., 2010a, 2010b).

The railing system along the canals will function as the foundation for the resource gathering within the project, but what if the system fails, needs maintenance, etc? If this happens the built in control system (part of the feedback loop) will take over by resolving to a secondary system that was developed along side the railing system. This other system is a series of easy to assemble and deployable floating solar stills that can be situated according to need, meaning if an area of the city is in desperate need of water, they will get a series of solar stills that can be hooked up to their local water network.



Figure 3. A section through a city, where a solar still, canals and roads are visible.

4.2. CITY COMPONENTS

As mentioned in the introduction to section 4, a city is composed of numerous different components, where the most important within this project is the water system. However of the multitude of different components of the project such as

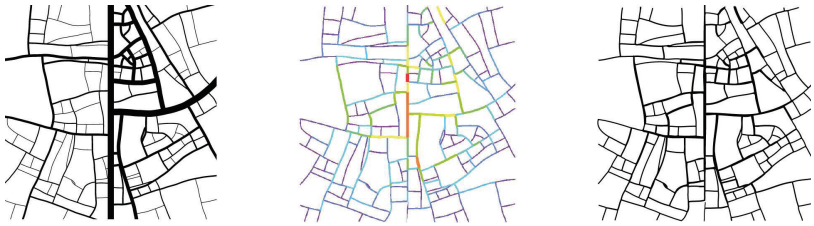


Figure 4. The optimization process of the road network, from left to right: original, data sheet and adapted outcome.

roads, electrical and distribution networks, this section will focus on the development of the road network.

The road was developed using a generative algorithm that is loosely based on the concept of desiccation (mud-cracking), where the size and time to break is a derivative of the metabolic scaling formula (2). However the interesting aspect of the road design isn't how they where made, but how the size of the roads was evaluated and readjusted according to its projected traffic load. The intention of the roads was that they should be metabolically flexible, meaning they should have the possibility to resize to every metabolically scaled size. Using an A* evaluation algorithm enabled the roads to do so. The A* algorithm makes every node in a network, search for the shortest path to every other node in the network (excluding themselves), as a path is "crossed" it gets counted and at the end of the process, every node and path has an associated count. This crossing count data can be used to determine if a road is appropriately scaled or needs to be resized to fit its projected traffic load (see images below) (Hart et al., 1968).

5. Evaluation

In order to determine whether or not there are being made positive changes to the city as a system, there must be an evaluation in correlation to the earlier stated ambitions – to create a system that can generate cities over time as a result of its internally generated resources. Following will describe two different means of evaluation, one being the use of a fitness for the cities and how this was determined and used. The other evaluation method sampled different patches through out the city proposals, and compared them to patches from real cities of the region. The different types of evaluation served different purposes, the fitness evaluation was to indicate which system had the highest functionality. The comparative analysis was to analyse how close the generated outcomes would be to cities that had been developed over a long period of time, rather then what we see now where western ideals are being implemented in a middle-eastern context.

5.1. THE FITNESS FORMULA

The fitness of the cities was calculated based on a formula that utilized the values in the table below (the table represents values from two generated cities).

$$\text{Fitness} = (\text{length of water network} / \text{number of buildings}) / \text{combined connectivity of the road and water networks.} \quad (4)$$

Dividing the canal network length by the number of buildings, gives a ratio as to how much canal there is per building, thereby giving a value as to how well the system have dealt with the issue of water access. The combined connectedness of the two networks is calculated by the A* algorithm (described in section 4.2). This portion of the equation is preferential to low numbers meaning that the connectedness of the roads and canals are high. The formula favours cities with a high number for the building to canal ratio and a high connectivity. Below are two images of some of the generated cities.

Table 2. Table displaying gathered values that were used to calculate the fitness of the cities.

	Number of buildings	**Length of water canals (m)**	**Sustainable population**	**AVG connectivity water**	**AVG connectivity roads**
Low-rise city	14.880	233.351,812	764.489	5.745	5.694
Tent Pole City	7.391	150.853,117	494.213	9.206	5.206



Figure 5. (left) Low-rise – City with a fitness value of 1,370, and (right) Tent Pole City with a fitness value of 1,410.

5.2. COMPARATIVE ANALYSIS

It was chosen to take samples of 5 by 1 kilometre from three vernacular cities within Qatar's climactic region, which was thought to represent a wide variety of

city morphology. Within the big samples smaller samples were made that were analysed for smallest and largest footprint, number of neighbouring houses, number of courtyards, to name a few. The cities that were tested were Bandar E-Abbas in Iran, Jeddah in Saudi-Arabia and Baghdad in Iraq. The same type of test was conducted on the generated city proposals, one of which is displayed bellow. The proposal with the lowest deviation was the best in terms of creating the "image of a city".



Figure 6. The sample to the left is from Jeddah, Saudi-Arabia, the other samples are from a city proposal. See table 2 for deviation data.

Table 3. Deviation data for figure 6.

	Mean area deviation	**Built area deviation**	**Unbuilt area deviation**	**Built % deviation**	**Mean neighbouring deviation**	**Average deviation**
Figure 6	0,969	0,675	1,225	1,407	2	1,259

6. Conclusion

The experience that was gathered using the city-generating algorithm led to the conclusion that cities, as dynamic and difficult to understand as they are, can, to an extent, be grown and designed holistically. In the near future, when this kind of design challenge becomes more than a project experiment, it is imperative that some of the logics and methods that have been developed in this project should be elaborated and implemented for the use in real-world applications.

It was learned that it is possible to create a mathematical growth model that revolves around the resources of a city. In order to prevent over or under production of architecture, it is seen as a possibility to embed this new model into future urban plans and/or cities. This is not saying that architecture does not posses the ability to promote change; there is simply a need for a new development paradigm that meets the needs of a city, but does not over interpret them. The remaining question is; did

the project produce a new methodology for urban development? In short, yes. Though it is not complete, the project proposes some innovative approaches towards formalizing, developing and implementing a new methodology for the urban places of the future. This project proposes that a resource driven urban model is a definite possibility, and should be realized.

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